Similarly, the kinetic energy per unit volume

$$= \frac{KE}{\text{volume}} = \frac{\frac{1}{2}mv^2}{V} = \frac{1}{2}\left(\frac{m}{V}\right)v^2 = \frac{1}{2}\rho v^2$$

ii) **Potential energy:** The potential energy of a liquid of mass m at a height h above the ground level is given by

$$PE = mgh$$

The potential energy per unit mass

$$= \frac{PE}{m} = \frac{mgh}{m} = gh$$

Similarly, the potential energy per unit volume $= \frac{PE}{\text{volume}} = \frac{mgh}{V} = \left(\frac{m}{V}\right)gh = \rho gh$

iii) **Pressure energy:** The energy acquired by a fluid by applying pressure on the fluid. We know that

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \Rightarrow \text{Force} = \text{Pressure} \times \text{Area}$$

$$F \times d = (P\,A) \times d = P\,(A \times d)$$

$$\Rightarrow F \times d = \text{W} = \text{P V} = \text{pressure energy}$$

Therefore, pressure energy, $E_p = PV$

The pressure energy per unit mass =

$$\frac{E_p}{m} = \frac{\text{PV}}{m} = \frac{\text{P}}{\frac{\text{m}}{\text{V}}} = \frac{\text{P}}{\rho}$$

Similarly, the pressure energy per unit volume $= \frac{E_p}{\text{volume}} = \frac{PV}{V} = P$

7.6.3 Bernoulli's theorem and its applications

In 1738, the Swiss scientist Daniel Bernoulli developed a relationship for the flow of fluid through a pipe of varying cross section. He proposed a theorem for the streamline flow of a liquid based on the law of conservation of energy.

Bernoulli's theorem

According to Bernoulli's theorem, the sum of pressure energy, kinetic energy, and potential energy per unit mass of an incompressible, non-viscous fluid in a streamlined flow remains a constant. Mathematically,

$$\frac{P}{\rho} + \frac{1}{2}v^2 + \text{gh} = \text{constant} \qquad (7.36)$$

This is known as Bernoulli's equation.

Proof:

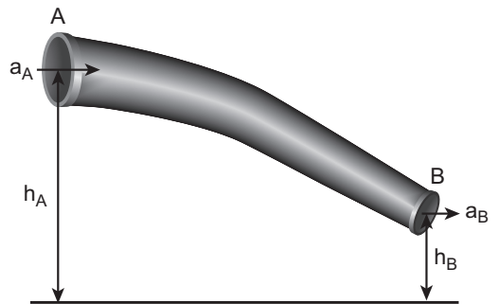


Figure 7.33 Flow of liquid through a pipe AB

Let us consider a flow of liquid through a pipe AB as shown in Figure 7.33. Let V be the volume of the liquid when it enters A in a time t which is equal to the volume of the liquid leaving B in the same time. Let a_A, v_A and P_A be the area of cross section of the tube, velocity of the liquid and pressure exerted by the liquid at A respectively.

Let the force exerted by the liquid at A is

$$F_A = P_A a_A$$

Distance travelled by the liquid in time t is

$$d = v_A\, t$$

Therefore, the work done is

$$W = F_A d = P_A a_A v_A\, t$$

But $a_A v_A t = a_A d = V$, volume of the liquid entering at A.

Thus, the work done is the pressure energy (at A), $W = F_A d = P_A V$

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